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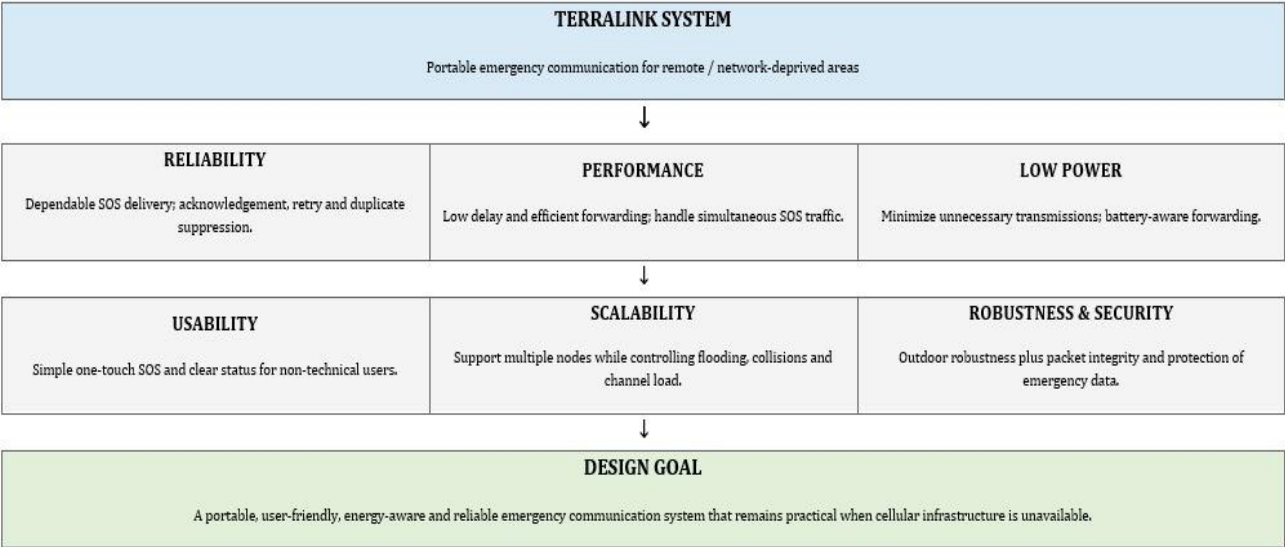
NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements define how well a system performs, focusing on core quality attributes such as **reliability, speed, scalability, and security**. They set the operational standards and constraints needed to ensure the system remains **practical, resilient, and efficient** under real-world conditions.

- **Reliability:** Guarantees SOS delivery using automated acknowledgements, retries, and duplicate filtering.
- **Performance:** Delivers low-latency packet forwarding while smoothly managing simultaneous emergency traffic over LoRa.
- **Low Power:** Maximizes battery life through duty-cycle optimization and energy-aware mesh routing.
- **Usability:** Features simple one-touch SOS activation and clear status feedback for non-technical users.
- **Scalability:** Supports multi-node expansion while actively preventing network flooding and packet collisions.
- **Robustness & Security:** Offers rugged outdoor protection, payload encryption, and self-healing network rerouting.

TERRALINK – NON-FUNCTIONAL REQUIREMENTS

Phase-2 Requirement Flowchart / Emergency LoRa Communication System



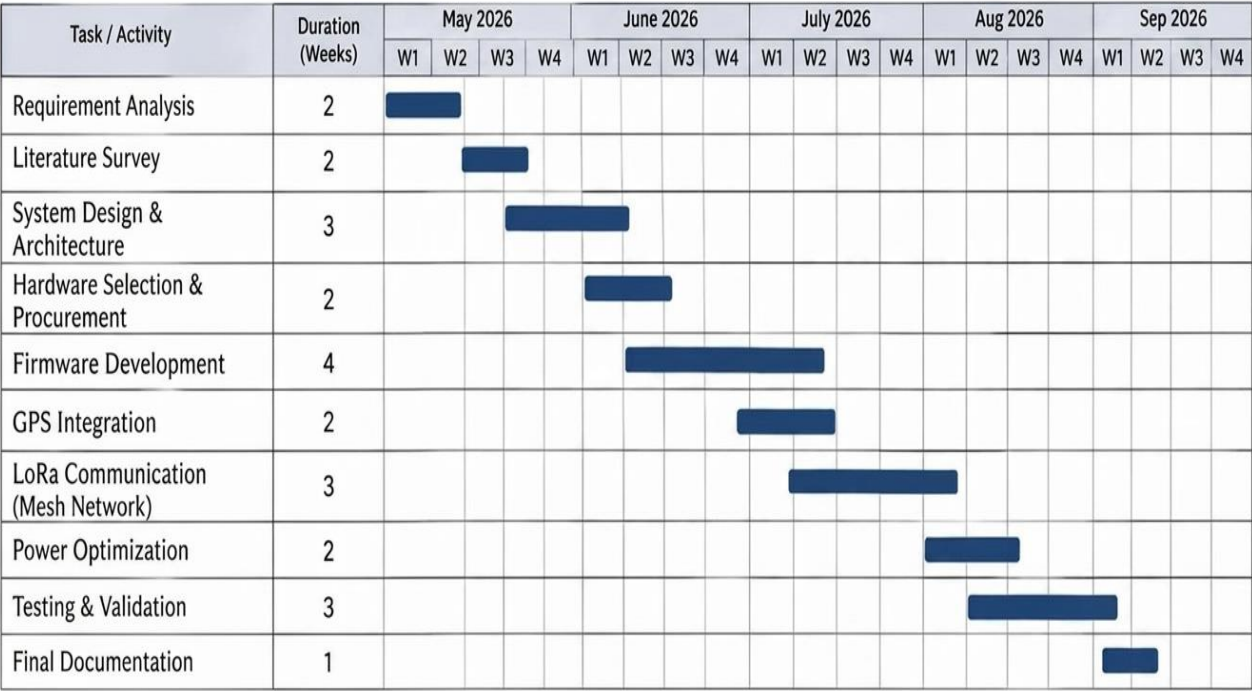
Functional requirements = WHAT the system does / Non-functional requirements = HOW WELL it should perform

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GANTT CHART

The Gantt chart shows the planned timeline and sequence of activities for the TerraLink project from May to September 2026. The project begins with requirement analysis and a literature survey, followed by system design, hardware selection, and firmware development. GPS integration and LoRa mesh communication are then implemented and tested. The project continues with power optimization, system testing, and validation. Finally, the project documentation is completed.

The major milestones include system design completion, development of a working LoRa mesh prototype, final testing, and project completion. This schedule helps the team manage time, track progress, and complete each stage of the project within the planned period.



Key Milestones

- System Design Complete – June 2026
- LoRa Mesh Working Prototype – July 2026
- Final Testing – Aug 2026
- Project Completion – Sep 2026

Legend:

- Task Duration
- Week